Macro-level interoperability in e-health

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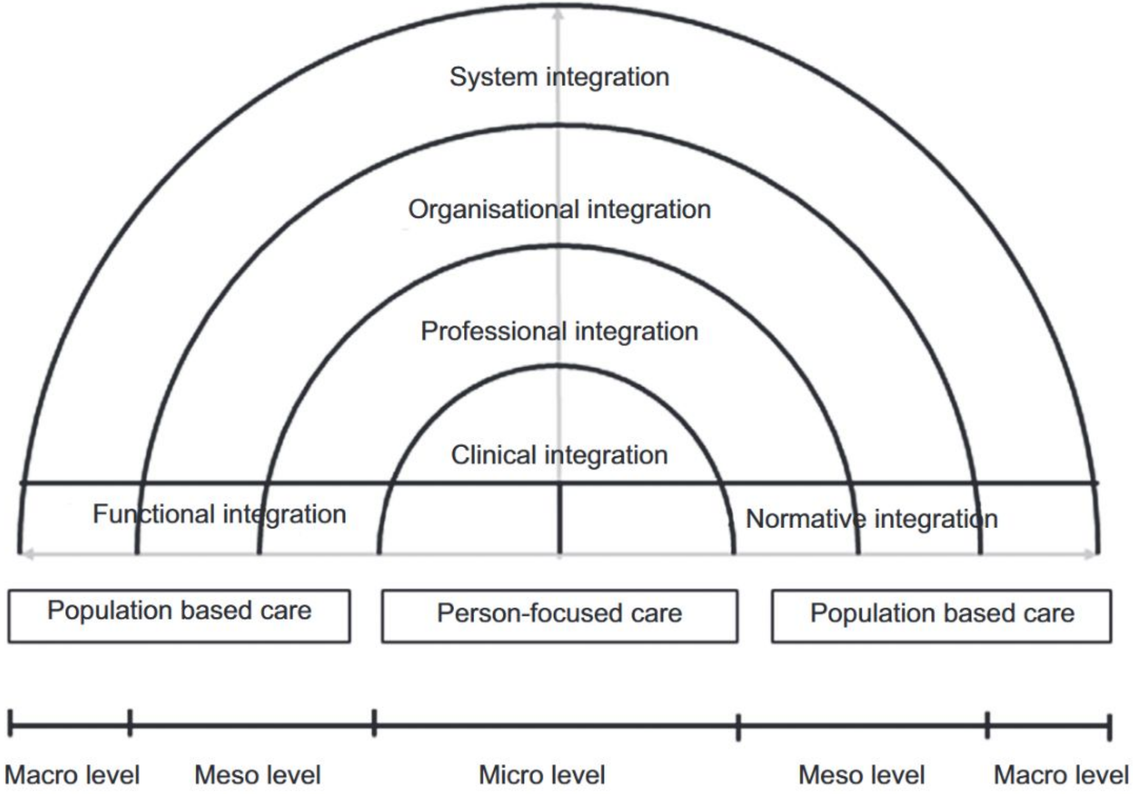
2023 - 2024	Digital Health Architect. National Health Data Analytics, Romania
2021 - 2023	E-health expert. Next Gen National EHR capabilities. Estonia
2019	Digital Health Architect. National Health Data Analytics, Kingdom of Saudi Arabia
2018 - 2019	Digital Health Architect. E-Health Architecture, MoH of Ukraine
2017 - 2020	Enterprise Architect. Health Sector Architecture, MoH of Mongolia
2016	Digital Health Architect. ITU Digital Health Acceleration Kit
2015	E-health Expert. Feasibility Study for Personalised Medicine in Estonia
2014 – 2018	Solution Architecture, Direct Access Laboratory Portal
2013 – 2014	Solution Architecture, X-Road pilot in Finland
2006 – 2011	Solution Architecture, Estonian National Health Records
1996 – 1999	Solution Architecture, Estonian Revenue Information System

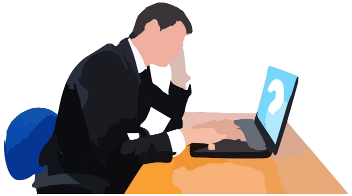
Integrated care

Integrated care is a complex social process influenced by macro-level circumstances such as legislative mandates, meso-level features such as organizational leadership, and micro-level interactions between health/social care providers and patients.



J. A. Shaw, P. Kontos, W. Martin, ja C. Victor, „The institutional logic of integrated care: an ethnography of patient transitions“, Journal o Health Organization and Management, kd 31, nr 1, lk 82–95, märts 2017, doi: 10.1108/JHOM-06-2016-0123

Martin discovers a need for personal prevention



Trigger

Martin reads a story about men's health from the newspaper. The story describes various health concerns that men of his age may face.
Martin decides to look into possible preventive activities.

Pre-visit

The family doctor's assistant asked Martin to fill out a personal health declaration before the doctor's appointment.
Martin filled out the health declaration via the patient portal.

Visit

During the appointment, the family doctor looked at Martin's medical history and asked additional questions about his lifestyle. The family doctor found no reason to refer Martin for further examination.

Self-help

Martin searched the web for healthcare services. A rich selection of laboratory tests along with nutritional counseling was offered in one location. The second site offered a variety of stress tests along with a personal training plan.
Martin was not sure if the services would fit his personal prevention plan.

Resignation

Since Martin didn't have more time to think about his personal prevention plan at the moment, he postponed the decision for the future.

Martin's moments of decision-making

- Do I need to make a personal health prevention plan now?
- Do I need to visit my GP for my personal prevention plan?
- Do I need to fill in a health declaration before the doctor's appointment?
- Should I declare having a good sleep, being a non-smoker, and having an active lifestyle?
- Do I need other healthcare services for my personal prevention planning?
- Does a web search result include useful references for my personal prevention plan?
- Do I need a lab test with a nutrition advisory for my personal prevention?
- Do I need a load test with a personal training advisory for my personal prevention?
- Can I postpone the personal prevention planning until I have more time?

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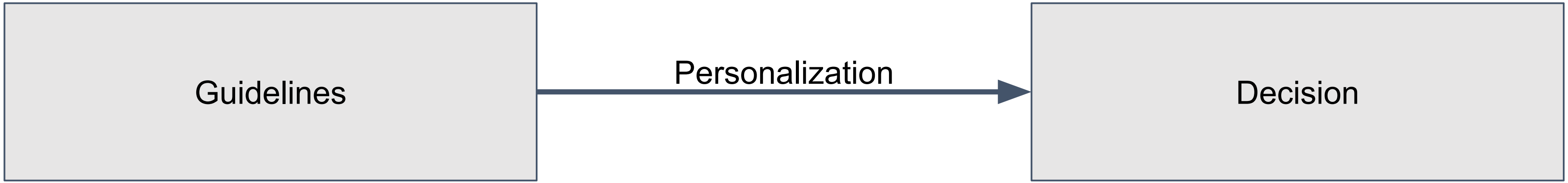
Martin decides to have his health status reviewed



men over 45, with an above normal body mass index, having history of heart issues in their family may need to have their health status reviewed and to consider revision of their lifestyle.

Name: Martin
Sex: male
Age: 51
Body weight: 94 kg
Body height: 182 cm
Family history: infarction
Medical history: outdated

Martin needs to have his health status reviewed

Analysis framework for the elements of decision-making (Zachman)

Who	Why	How	What	Where	When
person, role, organization, community, participant, stakeholder	purpose, mission, a choice of next action, an alternative way, a request, a task	logic, formula, algorithm, conversion, procedure, knowledge	Inputs, outputs, data, information, circumstances, condition, status	location, source or target system, physical or computer records, application	exact time, before or after an event, periodically, at a frequency, during an event

TAL TECH

https://www.sciencedirect.com/topics/computer-science/zachman-framework

Elements of Martin’s decision to look into his health prevention

- **Why**: to find out if the guideline from the newspaper matches his profile and decide about the personal prevention plan
- **How**: the guideline suggests that men over 45, with an above normal body mass index, having history of heart issues in their family may need to have their health status reviewed and to consider revision of their lifestyle.
- **Who:** home user / visitor / patient
- **What**: age, sex, body weight, body height, family history, medical records, and health goals.
- **Where**: personal memory, family member, patient portal.
- **When**: campaign in the news

https://www.sciencedirect.com/topics/computer-science/zachman-framework

Information used for decision-making

Decision (Why)	Information (What)
Do I need to make a personal health prevention plan now?	Age, sex, body weight, body height, family history, health records, lifestyle, and health goals
Do I need to visit my GP for my personal prevention plan?	habits, community advice
Do I need to fill in a health declaration before the doctor's appointment?	request by the receptionist
Should I declare having a good sleep, being a non-smoker, and having an active lifestyle?	personal observations according to his memory
Do I need other healthcare services for my personal prevention planning?	health concerns, health goals, health check summary document by GP
Does a web search result include useful references for my personal prevention plan?	web search results, personal assessment
Do I need a lab test with a nutrition advisory for my personal prevention?	health concerns, health goals

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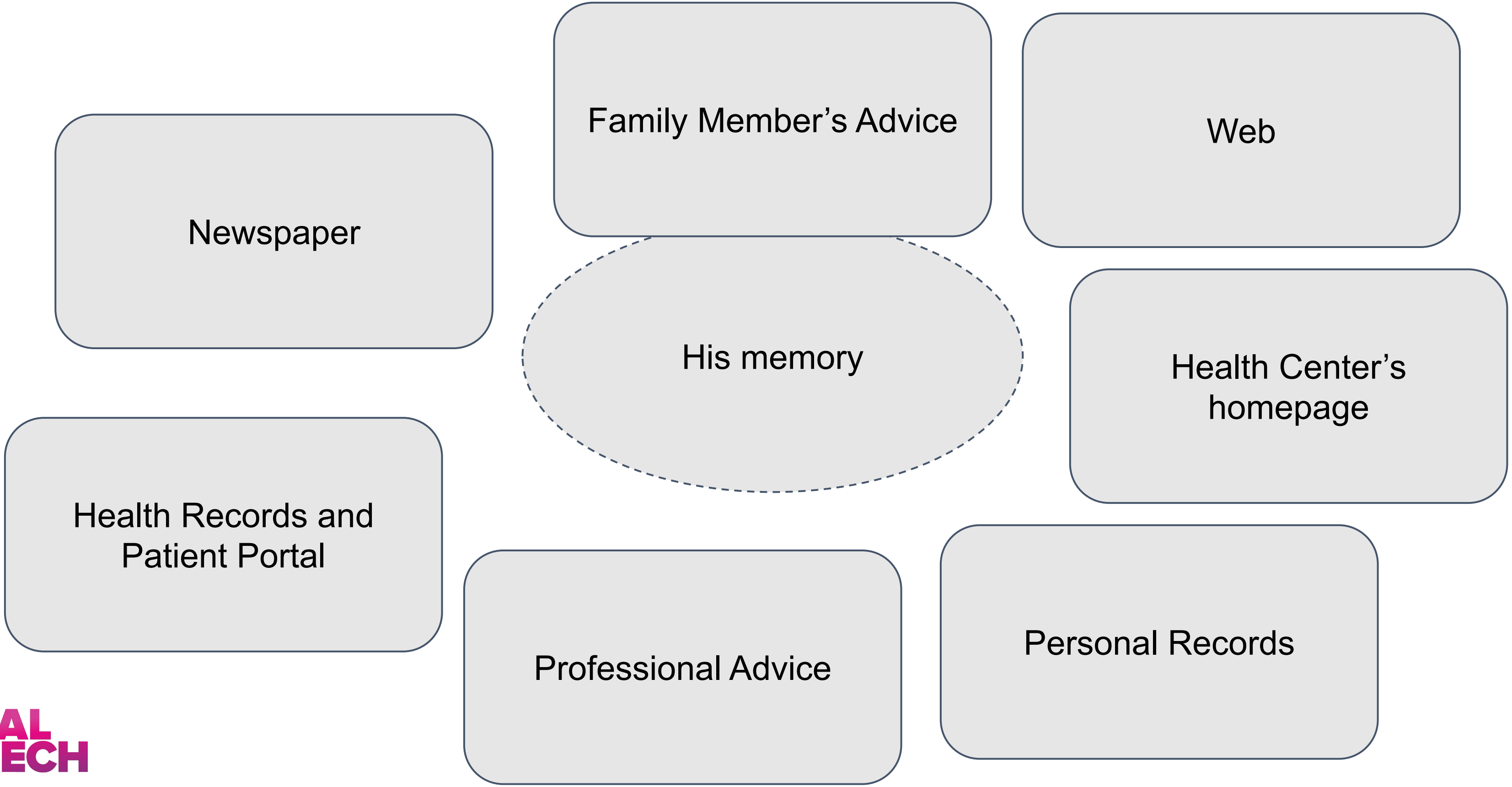
Martin’s systems of health information



GP conducts a health checkup for personal prevention

Trigger

A call of a patient

Pre-visit

Receptionist schedules an appointment with GP and recommends the patient to fill in his health declaration.

Exam

During the appointment, the family doctor looked at patient's medical history and asked additional questions about his lifestyle. The family doctor found no reason to refer the patient for further examination.

Advice

Patient asked for an advice about out-off-the-pocket health prevention services. GP was not able to suggest specific services or providers.

Passivation

The patient left the family medical center. The center does not plan for any further activities in relation to the patient.

GP's systems of health information

Reimbursement Claims Reporting

Guidelines & Handbooks

Public Health Reporting

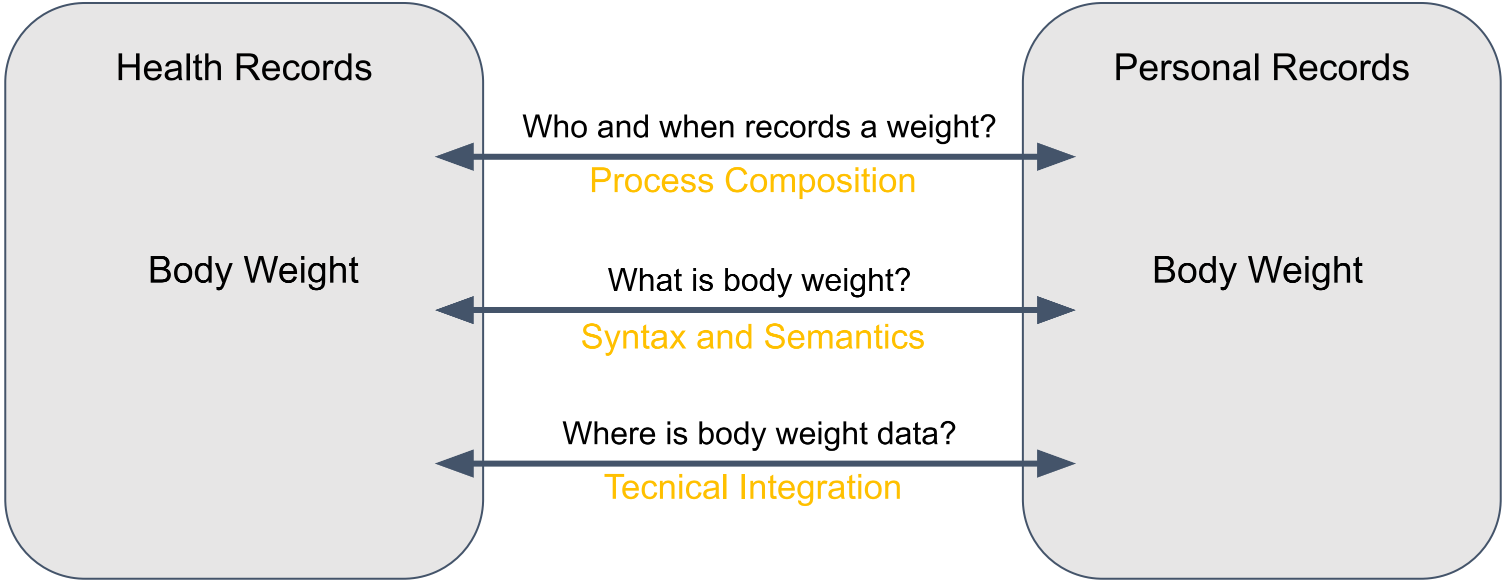
Patient's memory and body

Health Records

Case Report Forms

Medical Records

Can Martin share his personal records with a physician?



[1] A. Tolk, „Levels of Conceptual Interoperability“, 2003 Fall Simulation Interoperability Workshop.

[2] A. Tolk, L. J. Bair, ja S. Y. Diallo, „Supporting Network Enabled Capability by extending the Levels of Conceptual Interoperability Model to an interoperability maturity model“, Journal of Defense Modeling & Simulation, kd 10, nr 2, lk 145–160, apr 2013, doi: 10.1177/1548512911428457.

Interoperability of the health data

- **Technical Integration** - the sources of information are connected into a computer network and securely share data - data in the right location
  - Martin approaches all the sources separately and maintains their integrity manually
- **Syntax and Semantics** - the data is linked to shared terminologies and master data - data - data content fits the decision-algorithms
  - Martin's personal identifier, body weight, age, and sex have common meaning for both personal and health records
  - Martin's family history, lifestyle data, and health goals are unstructured data, which need human interpretation always
- **Process Composability** - the flow of data satisfies the needs of downstream decision-making - data is timely and actionable
  - Martin needs to recapture his family health history. Most probably his smoking status needs recapture for public health research.

A public health researcher monitors health risks of

Trigger	Sources	Analysis	Indicators	Reporting
Public health monitoring plan, extraordinary study of risk factors	The researcher considered the national health information system, the information systems of hospitals and family doctors, medical claims, interviewing people, clinical research, and disease registries as data sources.	For both studies, it was necessary to know the smoking habits; for the cardio risk factor study, it was also important to know the heart disease diagnoses of the respective people.	Indicators on the consumption of alcohol and tobacco products were part of the routine monitoring. In addition to the monitoring plan, the institute also planned an extraordinary study of risk factors for cardiovascular diseases.	annual public health monitoring, extraordinary study of risk factors

WHO List of 100 Core Health Indicators

- **Health status** indicators include core indicators including mortality by age, sex and cause (that includes the 16 mortality related indicators of the health and health related SDGs as well as core morbidity and fertility indicators).
- **Risk factors** indicators include those relating to nutrition, environmental, behavioural, injuries and violence.
- **Service coverage** indicators reflect priorities across the spectrum of health services including reproductive, maternal, newborn, child and adolescent, immunization, HIV, TB, malaria, neglected tropical diseases, noncommunicable diseases, mental health and substance abuse.
- **Health systems** indicators include indicators of health system inputs and outputs such as health facility density and distribution, health workforce, health information and quality and safety of care, health security capacity.

"Global Reference List of 100 Core Health Indicators," SCORE for health data. Accessed: Jul. 20, 2021. [Online]. Available: https://score.tools.who.int/tools/enable-data-use-for-policy-and-action/tool/global-reference-list-of-100-core-health-indicators-72/

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Tobacco use among persons aged 15+ years [SDG 3.a.1]

Definition: Age-standardized prevalence of current tobacco use among persons aged 15+ years.

"Smoked tobacco products" includes cigarettes, cigarillos, cigars, cheroots, bidis, pipes, shisha (water pipes), roll-your-own tobacco, kreket, and any other form of tobacco that is consumed by smoking.

"Smokeless tobacco" includes moist snuff, plug, creamy snuff, dry snuff, plug, dissolvables, gul, loose leaf, red tooth powder, snus, chimo, gutkha, khaini, gudakhu, zarda, quiwam, dohra, tuibur, nasway, naas/naswar, shammah, betel quid, toombak, pan (betel quid), iq'mik, mishri, tapkeer, tombol and any other tobacco product that is consumed by sniffing, holding in the mouth, or chewing.

"Current use" means use at the time of the survey, whether daily use or occasional use.

Numerator: Number of current tobacco users aged 15+ years.

Denominator: Total population aged 15+ years.

Disaggregation: additional dimension: Age, sex

Method: Number of respondents aged 15+ years currently using any tobacco product (smoked or smokeless)/(number of survey respondents aged 15+ years) x 100.

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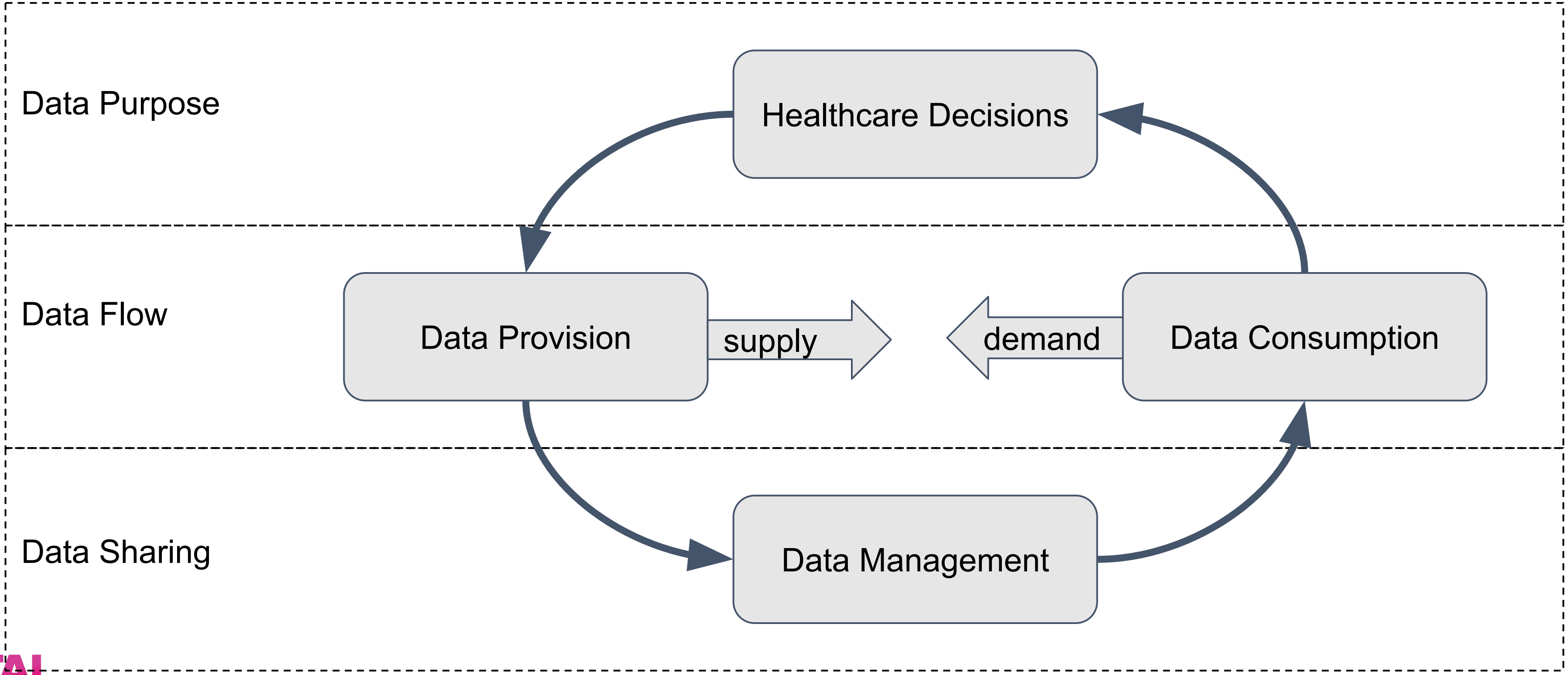
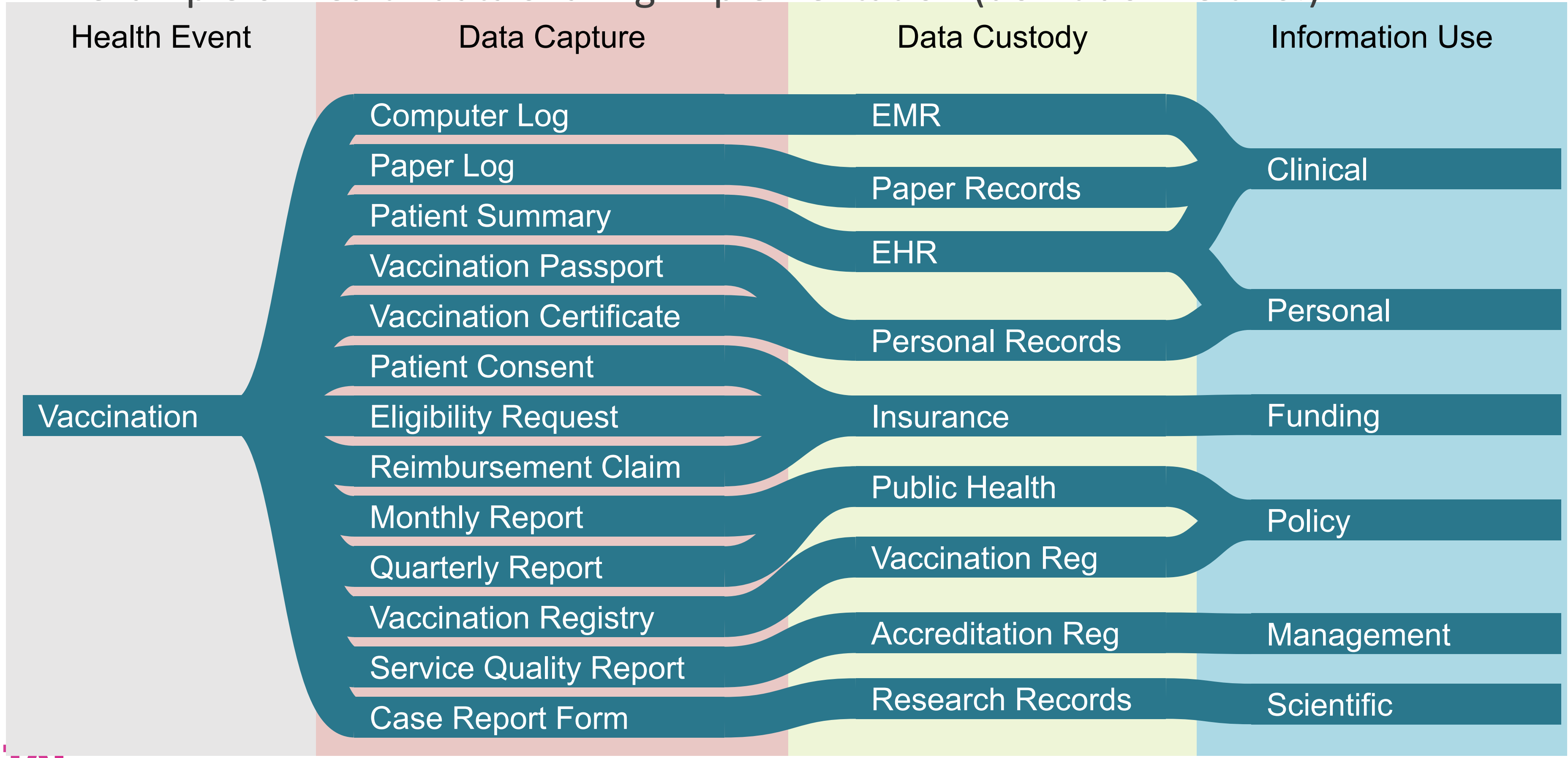
Tensions between supply and demand make data flow



An example of health data sharing implementation (don't do like this!)



Ministerial decree: Requirements for documentation of healthcare services

Issuer: Minister of Social Affairs

Type of Act: decree

Enforced date: 13.04.2025

Valid to: in force

https://www.riigiteataja.ee/akt/117082021010

IMMUNISEERIMISED

ajakava		kuupäev	seeria, doos	allkiri, kood
vanus	vaktsiin			
3-5 päeva	BCG 1			
3 kuud	PDT 1+OPV 1			
4,5 kuud	PDT 2+OPV 2			
6 kuud	PDT 3+OPV 3			
1 aasta	MMR 1			
2 aastat	PDT 4+OPV 4			
7 aastat	dT 5+OPV 5			
8 aastat	BCG 2			
12 aastat	dT 6			
13 aastat	MMR 2			
17 aastat	dT 7			

Plaanivälised immuniseerimised

vaktsiin	kuupäev	seeria, doos	allkiri, kood

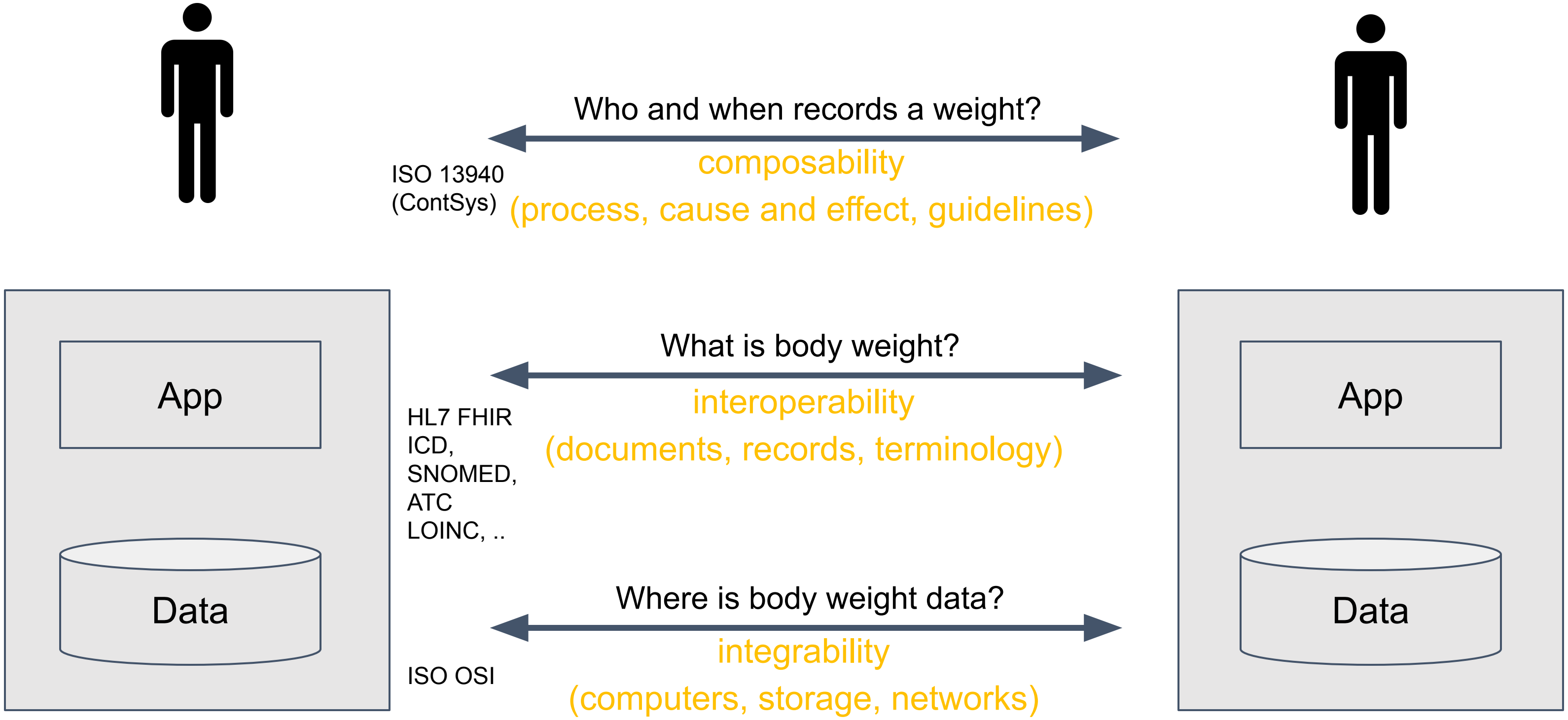
Kõrvaltoimed

Logic of Data Sharing

	Governance	Ecosystem agility	Data use for multipurpose	Data Standards
Informational (vision/ expectation)	Diverse (confederat., matrix)	High	Fundamental	Dynamic
Shared registries (state-of-art)	Monopoly, Fragmented, Federative	Low	Static	Central
Transactional (common)	Bilateral, Multilateral	Moderate	Secondary	Contractual

intended model

Levels of interoperability



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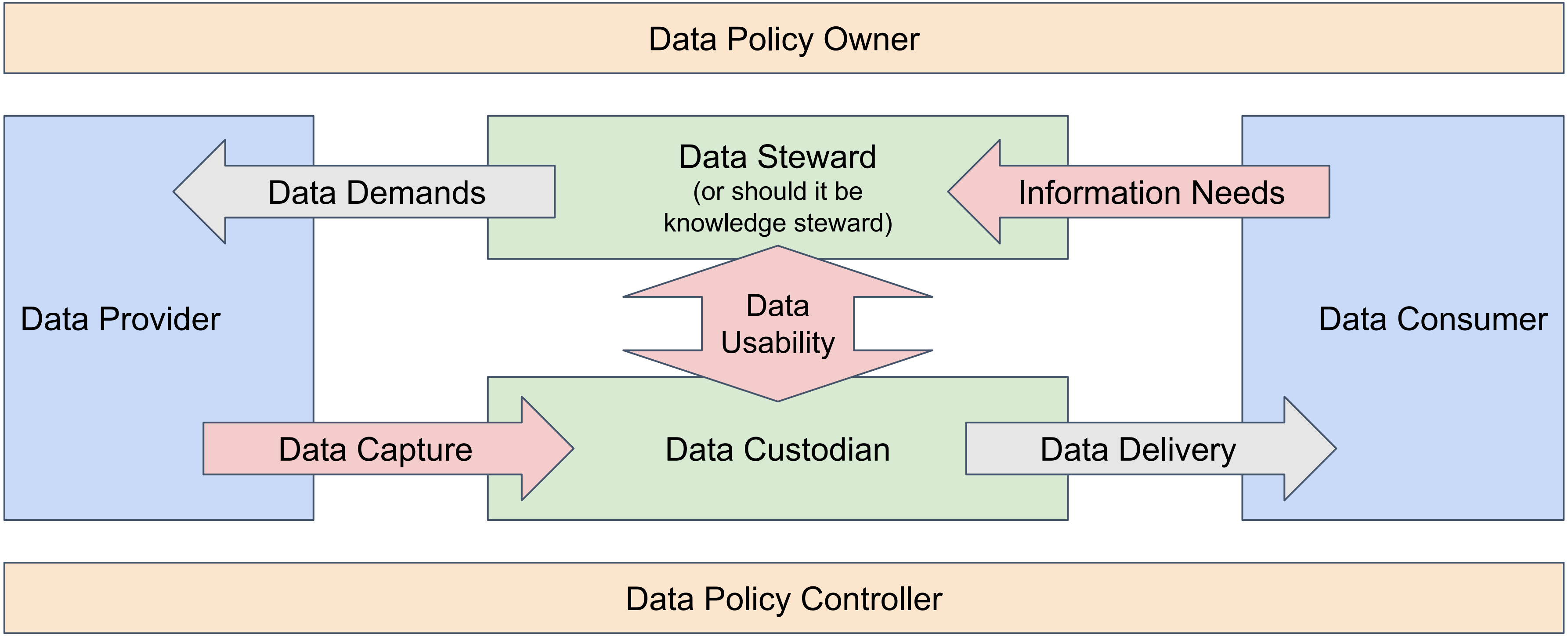
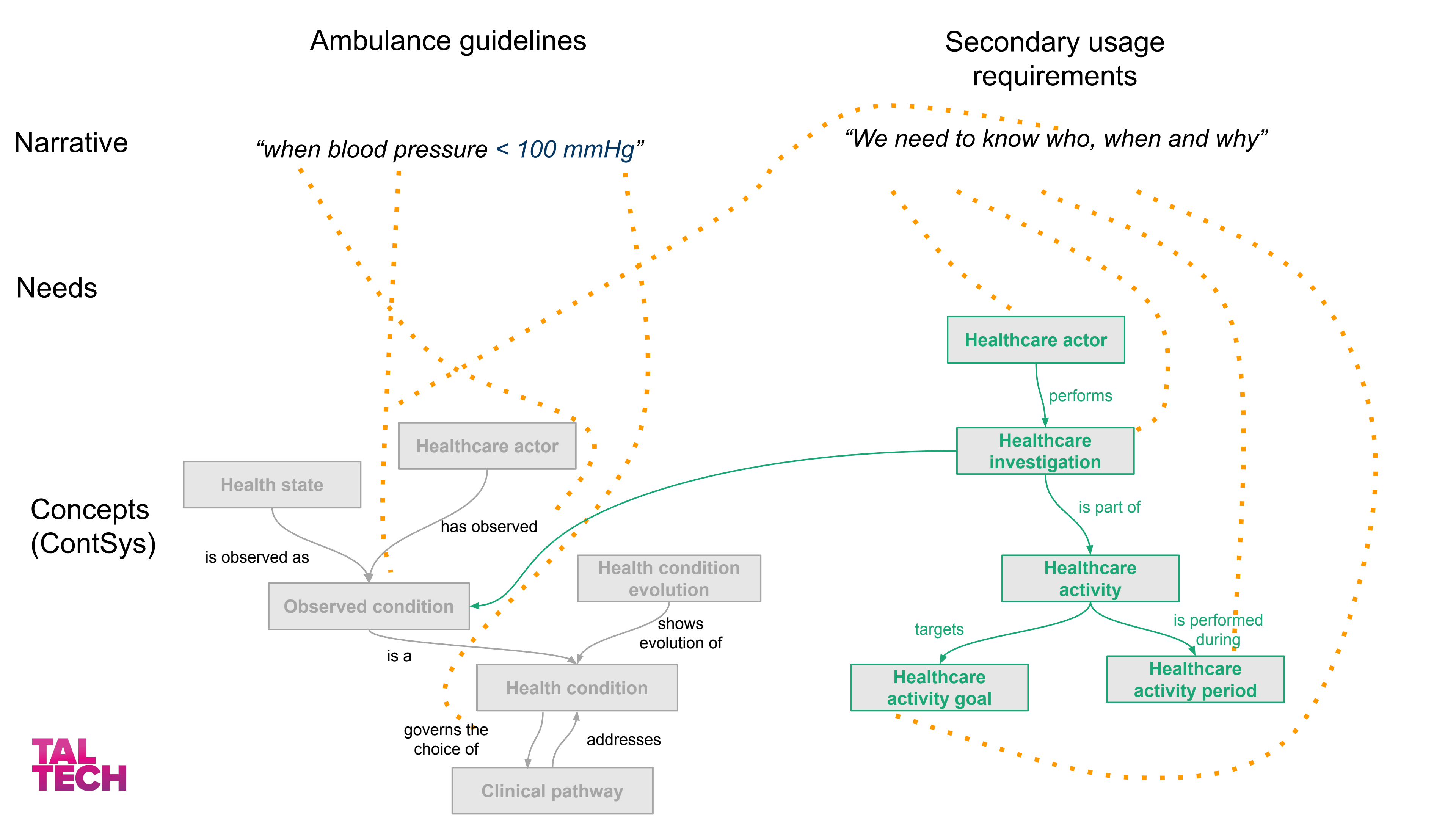
Governance concepts of data sharing





Ambulance guidelines
Secondary usage requirements
Narrative
Needs
Concepts (ContSys)
“when blood pressure < 100 mmHg”
“We need to know who, when and why”
Healthcare actor
Health state
Observed condition
Health condition evolution
Health condition
Clinical pathway
has observed
is observed as
is a
shows evolution of
governs the choice of
addresses
Healthcare actor
performs
Healthcare investigation
is part of
Healthcare activity
targets
is performed during
Healthcare activity goal
Healthcare activity period
TAL TECH

Health data policy shall empower pervasive data stewardship

Policy: data sharing regulations shall be harmonized against an explicit ontology (contsys)

Impact: Selection and Adaptation of a Concept System - Contsys ISO 13940 has been translated

Impact: Establishing the principle and responsibility of ontological compatibility of data requirements

Impact: Creation of a comprehensive management environment for data requirements

Impact: Creation of a tool for systematic validation of data requirements (ontological compatibility)

Impact: Linking data needs, information models, terminologies and message templates to validated requirements - work process and tool

Rationale: the continuity of health care processes (including the treatment process) is based on effective digitalization, the prerequisite for effective digitalization is multiple use of data, the prerequisite for multiple use of data is a compatible system of concepts

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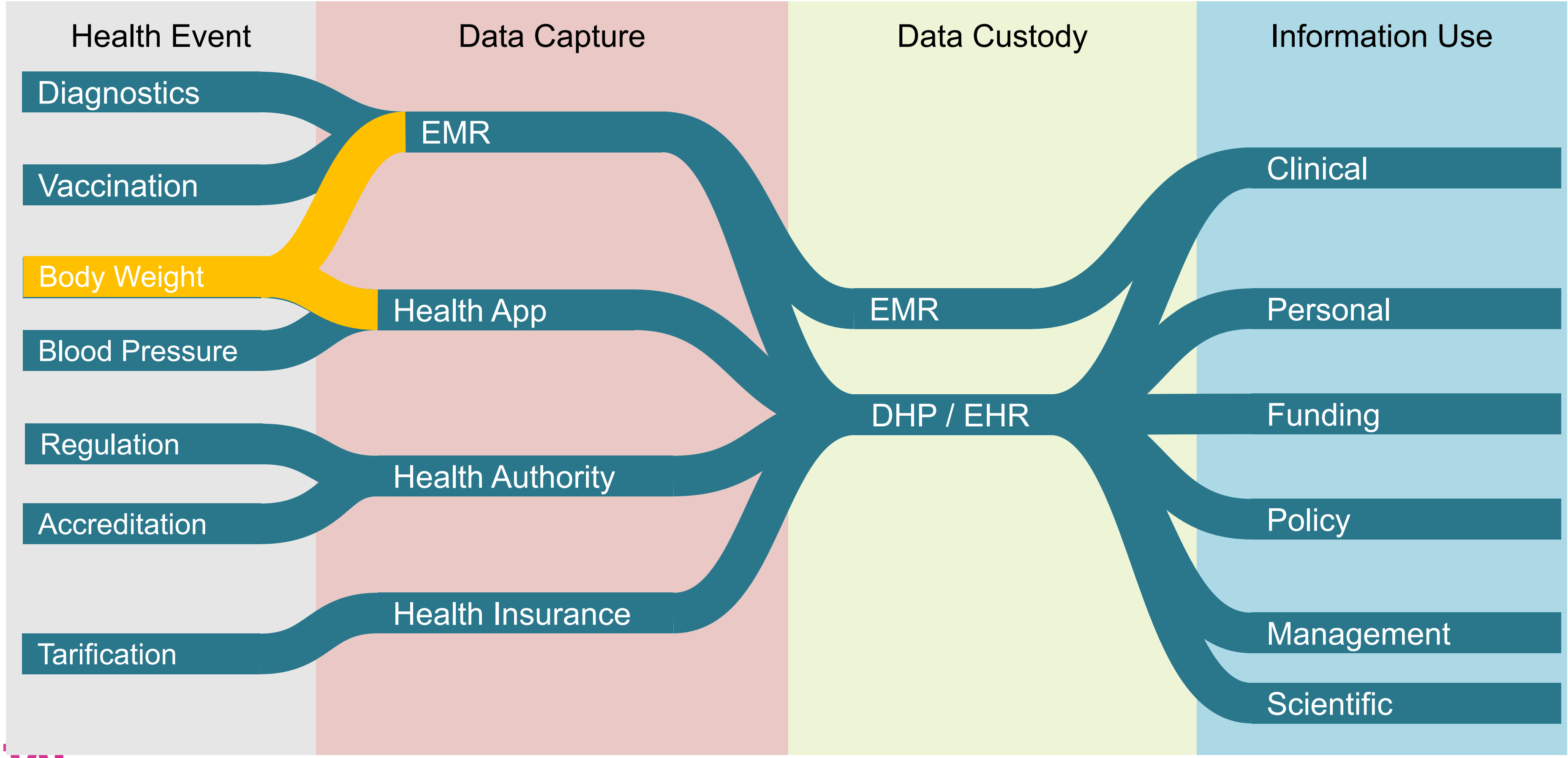
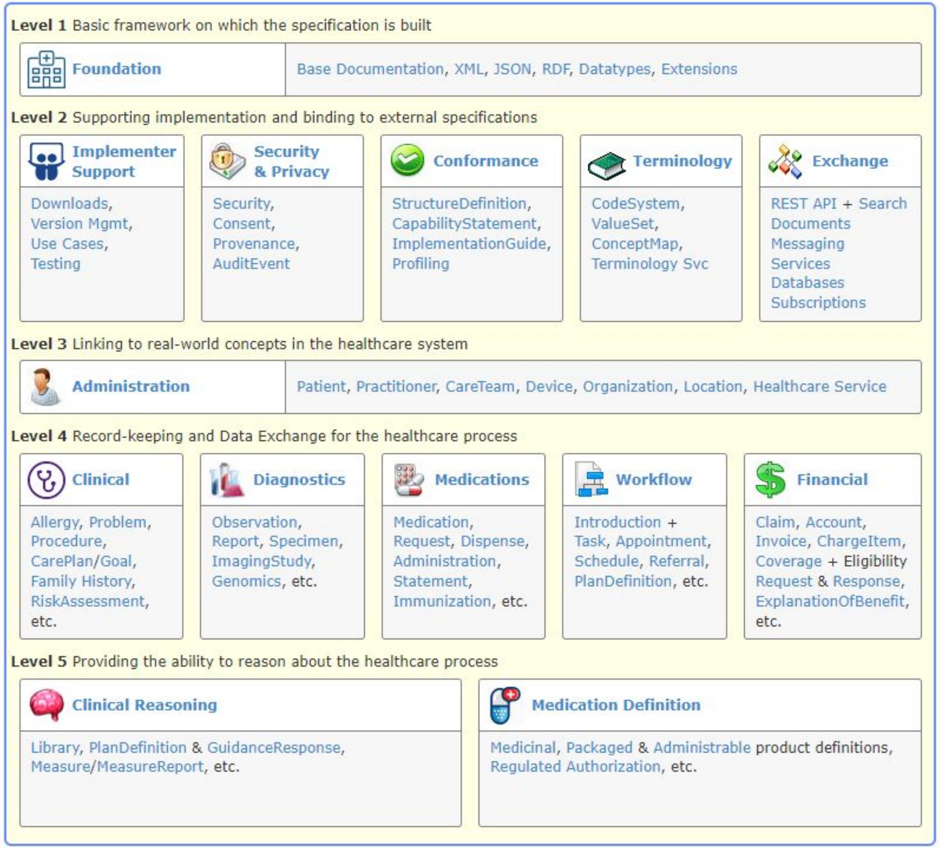
A vision of an ideal way of sharing data for multiple use (COUMT)



HL7 FHIR Levels



Level 1 - technologies for encoding data into **digital formats**; data types and base structures for utilizing computer networks

Level 2 - rules and structures for coding **metadata**; data that specifies security, conformance, terminology, and exchange features of a health data space

Level 3 - rules and structures for **master data** management; data that enables linking of records to people, organizations, devices, locations and services

Level 4 - rules and structures for **operational data** management; data that records clinical, diagnostics, medication, workflow, and financial events

Level 5 - rules and structures for **knowledge data** management; digital knowledge sharing allows rapid update of the algorithms supporting clinical workflow or medication guidance

https://hl7.org/fhir

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Information flows / ehealth capabilities

- Clinical documentation sharing
- Diagnostic image sharing
- Laboratory orders and reports
- Insurance claims reimbursement
- Public health reporting
- Prescription to drug-drug-interaction decision support
- Appointment booking
- Rescue service to ambulance to ER
- Clinical trials hires/reports/monitoring
- Professionals registration
- Healthcare providers registration
- Home monitoring
- Self-triage to referral
- Video consultation
- Digital intervention apps

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Health Information Sharing Capability FW

Governance Alignment

- Stakeholder Governance
- Sustainability Governance
- Data Governance

Process Alignment: composability of information and action

- Usability and Workflow
- Activity Alignment
- Participation
- Consent & Privacy

Technical Alignment: interoperability and integration of data

- Data Quality
- Data Security
- Data Transport
- Data Interoperability

1. Health Information Sharing Maturity Model. MITRE Health. https://health.mitre.org/blog/health-information-sharing-maturity-model-2/

Health Information Sharing Maturity Assessment 2020

		1 Empowering	2 Clinical	3 Management	4 Research	5 Policy	6 Funding
Technical	01 Data quality	2	2	2	3	2	3
	02 Data transport	2	3	2	2	2	3
	03 Data security	4	4	4	4	4	4
	04 Interoperability	3	3	3	2	2	3
Process	05 Usability	3	3	3	3	2	3
	06 Alignment	2	3	3	3	2	2
	07 Participation	4	3	4	3	3	2
	08 Consent	3	3	3	3	2	2
Governance	09 Data governance	3	3	2	2	2	2
	10 Stakeholder governance	2	3	3	2	2	2
	11 Sustainability	4	2	2	4	3	4

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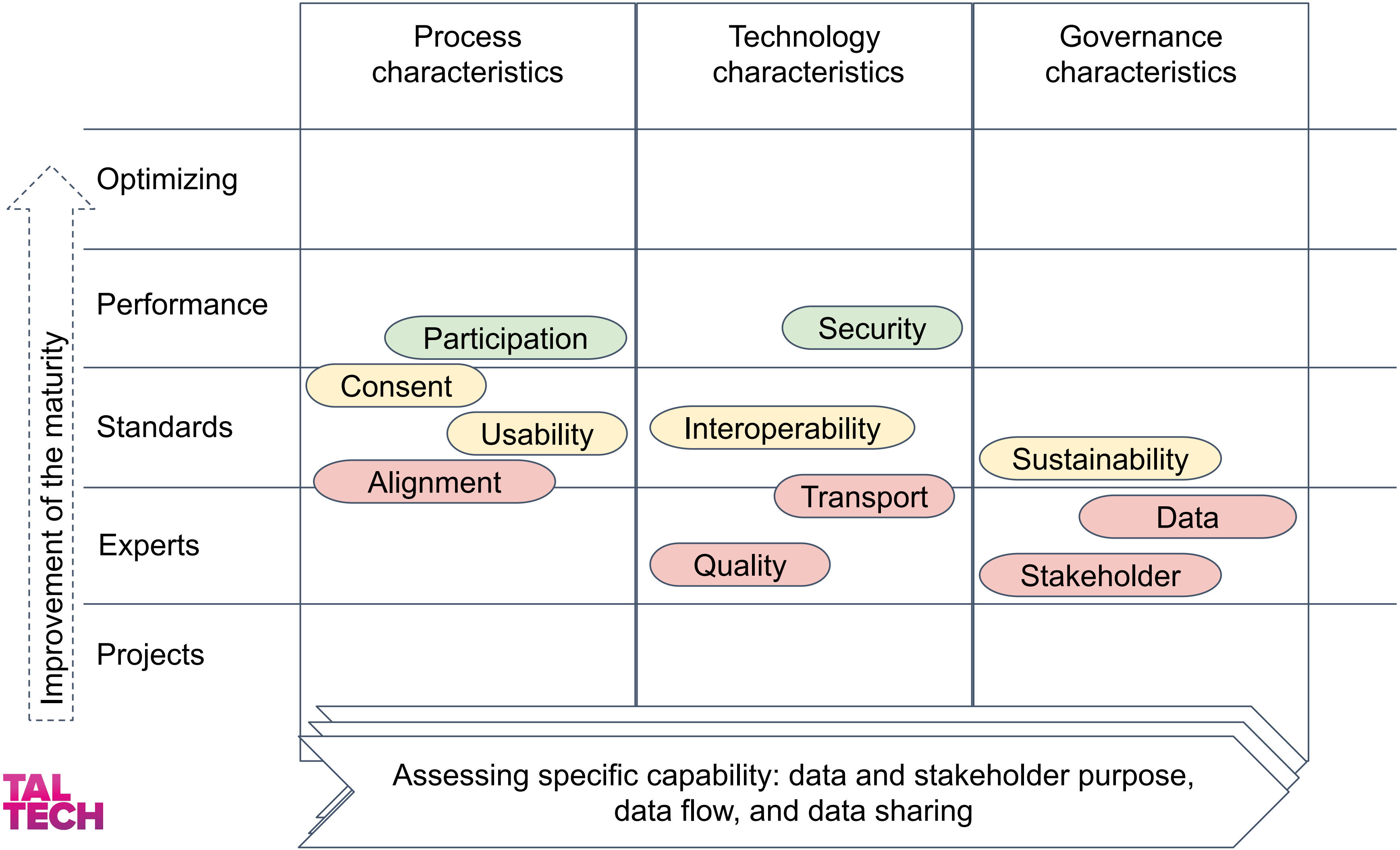

Process characteristics
Technology characteristics
Governance characteristics
Optimizing
Performance
Standards
Experts
Projects
Improvement of the maturity
Participation
Security
Consent
Usability
Interoperability
Sustainability
Alignment
Transport
Data
Quality
Stakeholder
Assessing specific capability: data and stakeholder purpose, data flow, and data sharing

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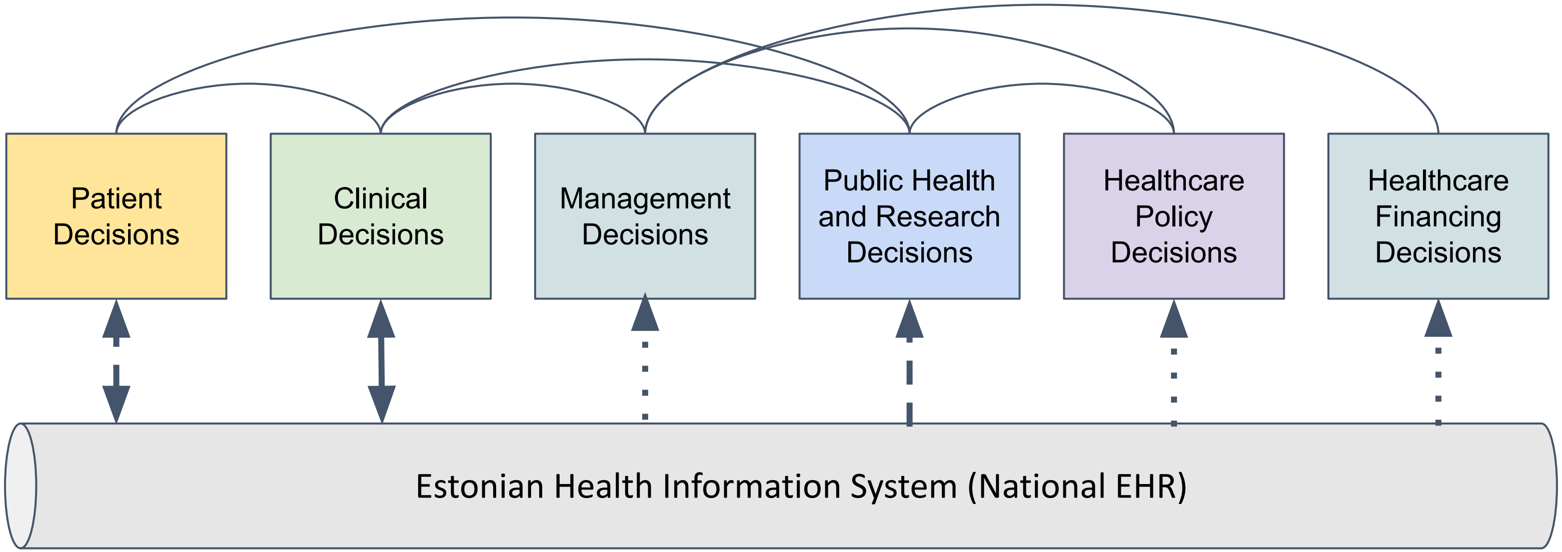
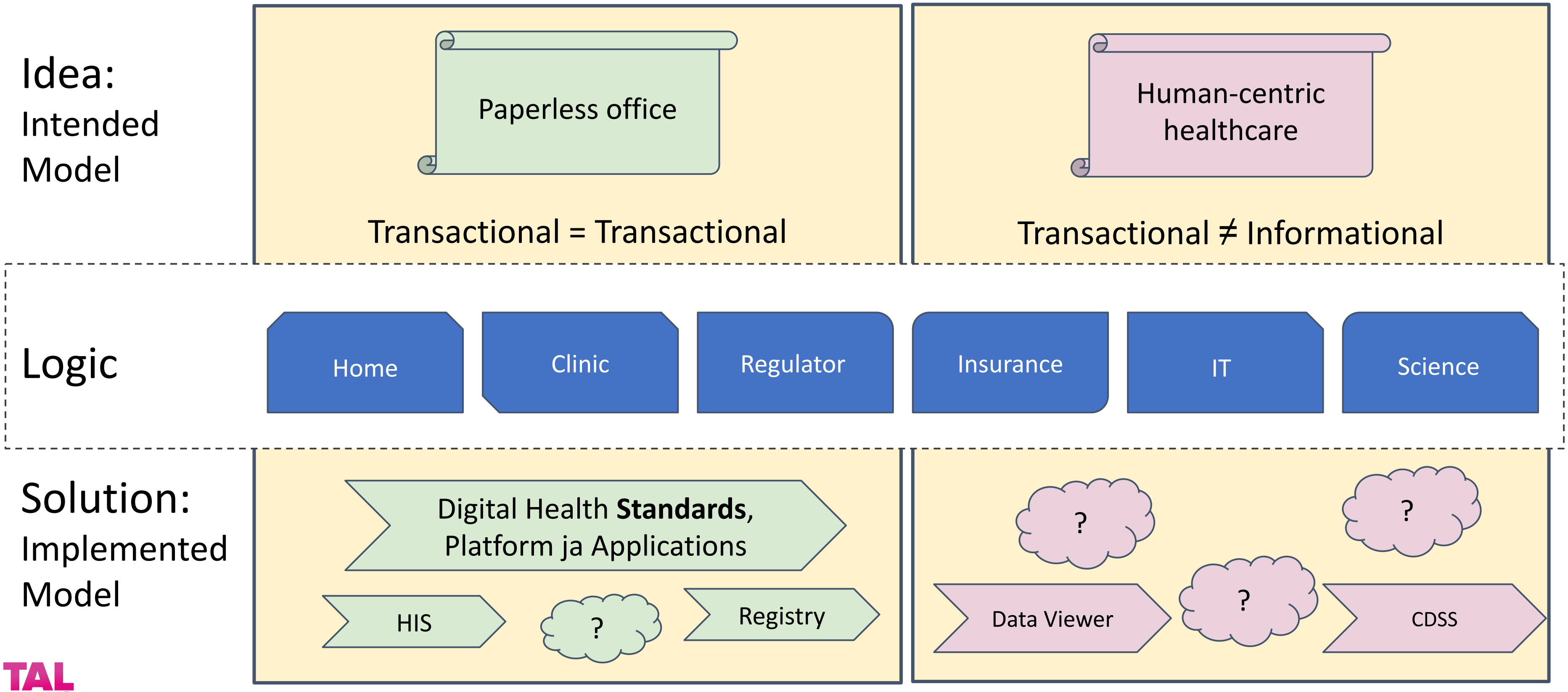
Findings: Still lots of point-to-point data exchange

Often the existing model of data exchange requires manual re-capture because of the interoperability issues



Institutional logic: implementation is based on technology and culture



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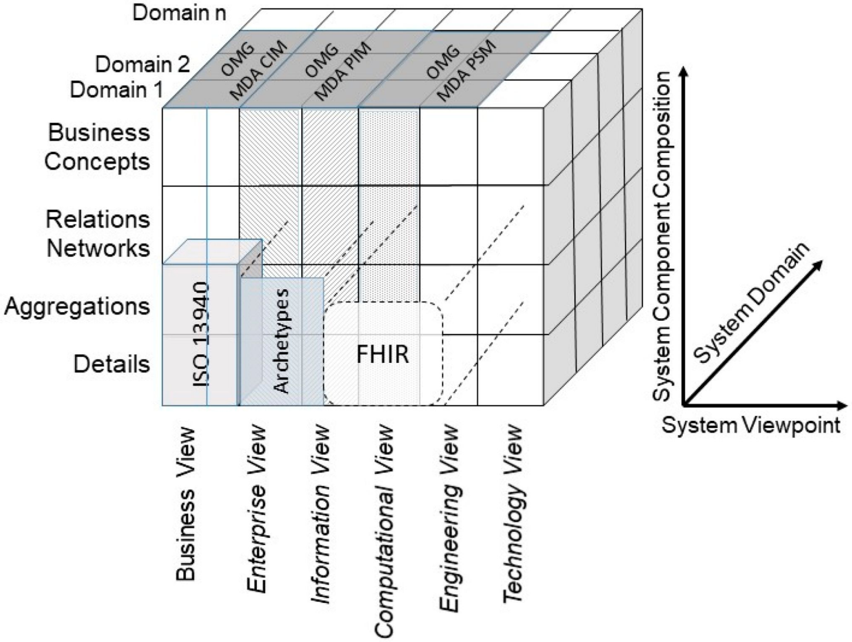
Making Sense of eHealth 3D - ISO 23903

ISO 23903 - Health informatics. Interoperability and integration reference architecture. Model and framework

- Viewpoint
  - Requirements, organization, tools, technology
- Components / Composition
  - Concepts, relations, details
- Domains
  - Capability, purpose, provision, custody, consumption



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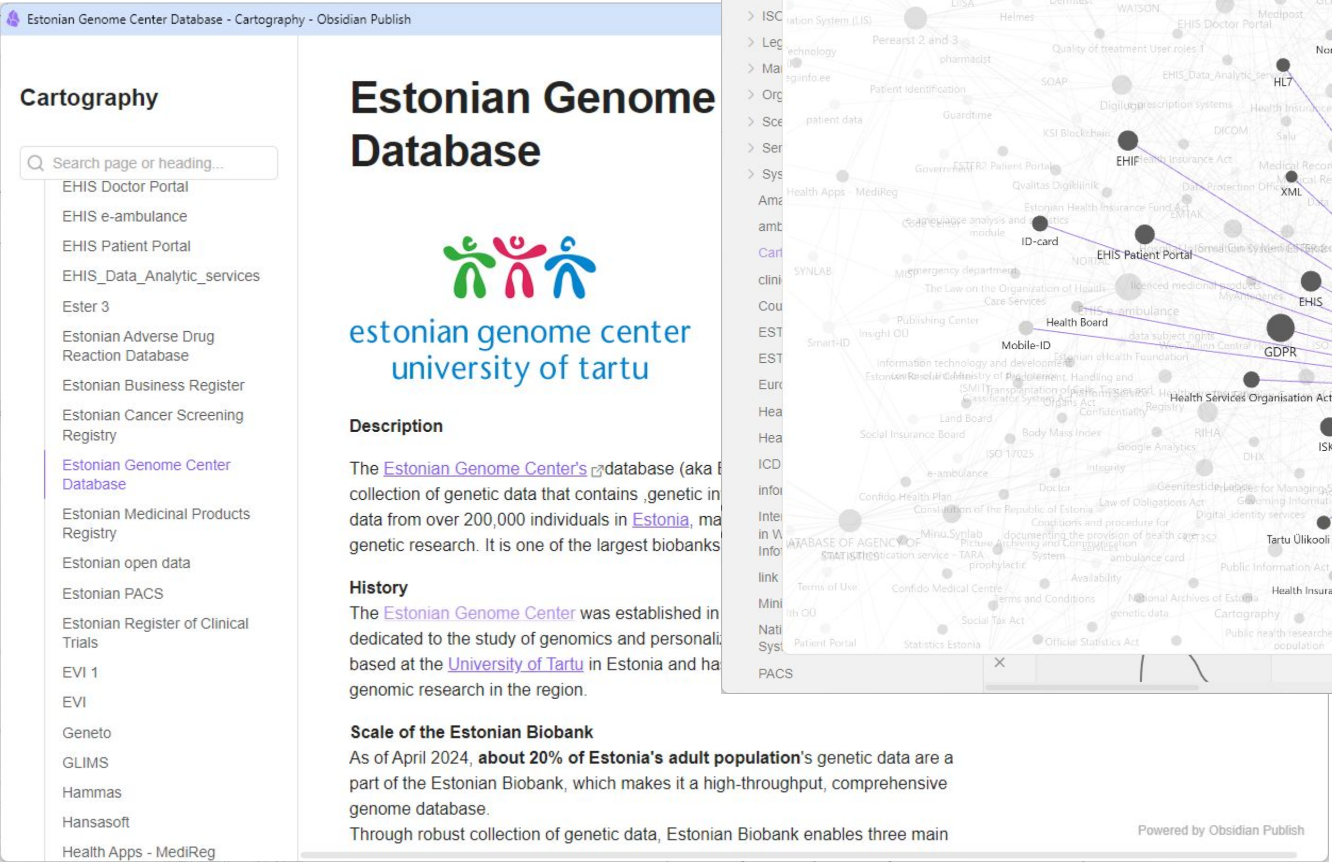
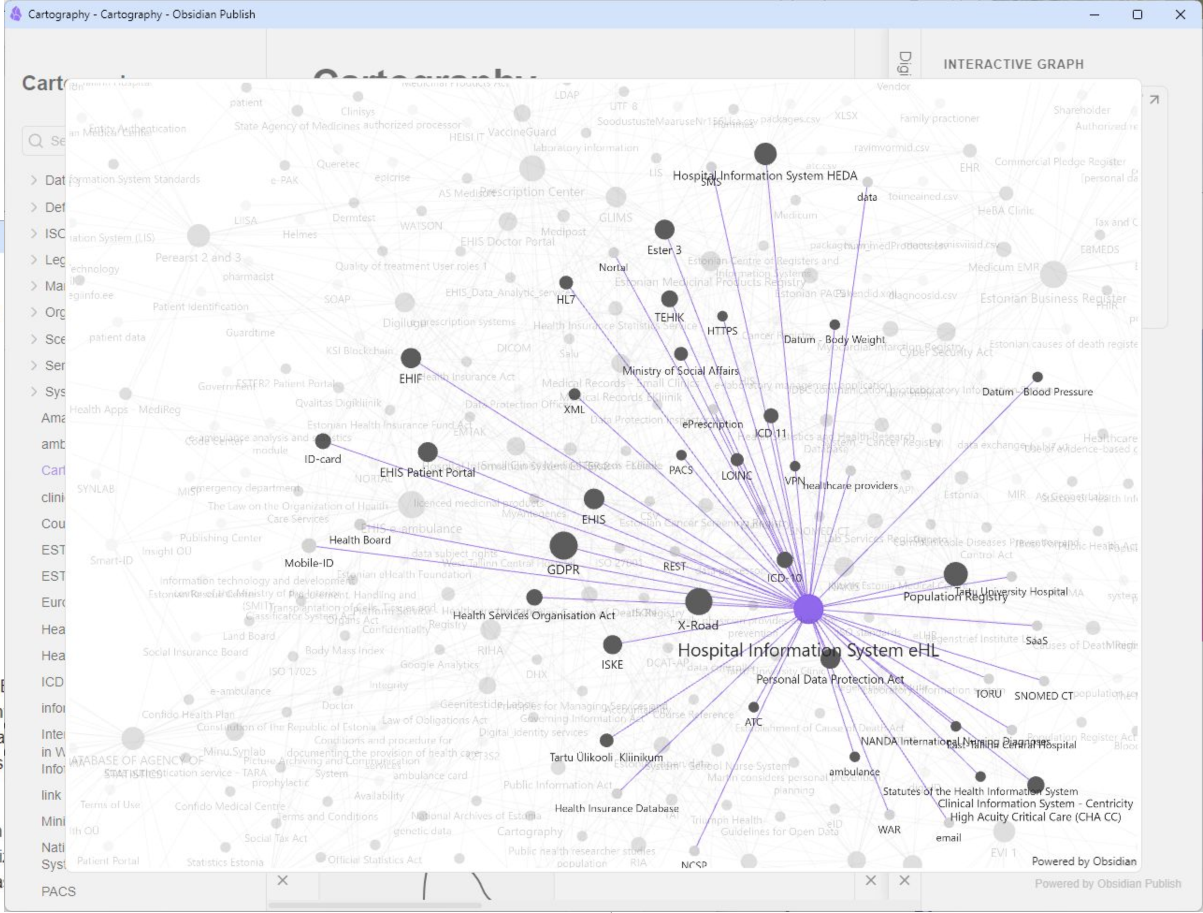
Cartography exercise

https://publish.obsidian.md/healthsense





Cartography - why, what, who, where

- **WHY** - propose a vision for one
  - Personal health journey
  - Medication and prescriptions
  - Sclerosis-multiplex research
  - Hospital network capacity
  - Lifestyle promotion
  - Health diagnostics

Contextualize

WHO participates

WHAT decisions

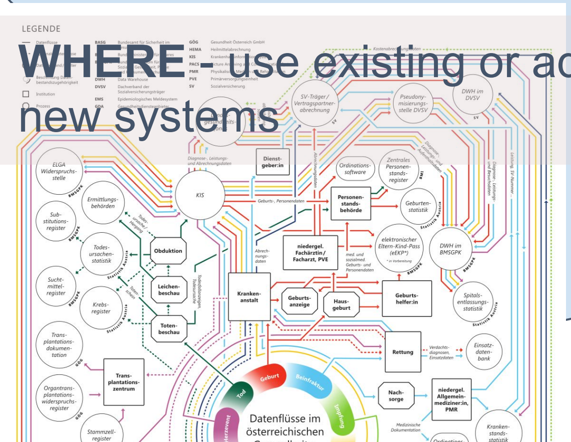
Austrian Health Data Space

Map

WHO uses

WHAT data

WHERE - use existing or add new systems



https://goeg.at/EU_Gesundheitsdatenraum_EHDS

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